Introduction to Statistical Decision Theory

Stats 220B

A note on 220A evaluations

Thank you for the **100% response rate** on the end-of-quarter evaluations.

That is rare and it matters a lot. It gives me a complete, unbiased picture of what worked and what did not.

220A went well overall. But there were consistent themes about things that could be better, and I want to address them head-on before we start.

What you asked for

1. **More worked examples, on the board.**
   Several of you said you learn best watching me solve problems step by step, in real time, rather than reading finished proofs off slides.
2. **Pacing: the first half moved slowly, the last week was a sprint.**
3. **Slide consistency: no multiple versions floating around.**

These are fair and actionable. Here is what changes.

What changes this quarter

- Slides carry definitions, theorem statements, and the logical backbone. But examples and proof sketches are worked **live on the board**, woven throughout every lecture.
- Pacing is even from Week 1 through Week 10. You have 220A behind you; the early weeks will not linger on prerequisites.
- One clean set of PDF notes posted after each lecture. One version. If I fix a typo, a single corrected file replaces the original.

Course information

STAT 220B: Statistical Decision Theory

Lecture: Monday & Wednesday, 11:00 AM – 12:20 PM, DBH 1300

Discussion: Wednesday, 12:30 – 1:20 PM, DBH 1300
Each week, 2 designated students will present a selected problem and help lead group discussion. I will moderate, step in when needed, and provide feedback.

Office hours:

- Instructor: 30 minutes per week (time TBD after consulting students)
- TA (Hanpu Shen): 1 hour per week (time TBD after consulting students)

Schedule note: next week

I will be serving on an NSF panel next week, so **both Monday and Wednesday lectures are cancelled**.

I do not want us to fall behind, so I would like to make up those two lectures. Two options:

1. **Afternoon makeup sessions:** Schedule both lectures in the afternoon next week (Monday and Wednesday), at a time that works for the group.
2. **Use discussion sections:** Use the first few Wednesday discussion sessions as makeup lectures instead.

Let me know which you prefer. I will send a poll after class, but feel free to speak up now.

Evaluations

Homework: 4 assignments, spread across the quarter.

Exams: Midterm 1 (Weeks 1–5), Midterm 2 (Weeks 6–9). Both held in class.

Grades:

- Homework: 30%
- Discussion presentation & engagement: 10%
- Midterm 1: 30%
- Midterm 2: 30%

Exam dates? Let's fix *NOW*!

Feedback policy

Student feedback is important for improving the course. A mid-quarter evaluation will be conducted to gather instructional feedback and help guide adjustments during the quarter.

To encourage participation in the official end-of-quarter course evaluation, the lowest-scoring completed homework will be dropped for all students who submit the evaluation.

References

Main text:

- Berger, *Statistical Decision Theory and Bayesian Analysis* (2nd ed., Springer, 1985)

Our path through Berger follows his "Mainly Decision Theory (Very Theoretical)" outline: Ch 1–2, selected sections of Ch 3–4, then Ch 5–8. Bayesian tools enter when they serve the decision theory, not as a standalone topic.

Secondary references:

- Ferguson, *Mathematical Statistics: A Decision Theoretic Approach* (1967)
- Lehmann & Casella, *Theory of Point Estimation* (2nd ed.)

Modern supplements:

- Efron, *Large-Scale Inference* (2010)
- Cesa-Bianchi & Lugosi, *Prediction, Learning, and Games* (2006)

Roadmap

Weeks 1–2: Decision problems, loss, risk, decision rules *Berger Ch 1–2*

Weeks 3–4: Bayesian decision theory, noninformative priors, admissibility of Bayes rules *Berger 3.3, 4.1–4.4, 4.8*

Week 5: Minimax theory, game theory, least favorable priors *Berger Ch 5*

Week 6: Minimax estimators, Stein estimation / **Midterm 1** *Berger 5.4–5.5*

Weeks 7–8: Invariance, equivariance, best invariant rules *Berger Ch 6*

Week 8–9: Sequential analysis (SPRT, minimax sequential) *Berger Ch 7*

Week 9: Complete class theorems *Berger Ch 8*

Week 10: Empirical Bayes, online learning / **Midterm 2** *Berger 4.5 + modern extensions*

Classical statistics vs. decision theory

Consider a drug company deciding whether to market a new pain reliever. Two unknowns matter: the proportion of people for whom the drug is effective (θ_1), and the market share it will capture (θ_2).

Classical statistics would use the data to make inferences about θ_1 and θ_2 – *without regard to the use to which they are to be put*.

Decision theory takes a different view. Beyond the sample data, two additional sources of information are relevant:

1. **Consequences of decisions:** quantified by a *loss function*. Overestimating θ_2 (overproduction) and underestimating it (missed sales) carry very different costs.
2. **Prior information:** knowledge about θ from past experience with similar drugs, similar markets, similar launches.

Decision theory attempts to combine all three – data, loss, and prior information – to make the best decision.

Why prior information cannot be ignored: Savage's example

Three experiments, all with *identical* statistical evidence:

1. A lady who adds milk to her tea claims she can tell whether tea or milk was poured first. In 10 trials, she is correct all 10 times.

2. A music expert claims he can distinguish a page of Haydn from a page of Mozart. In 10 trials, he is correct all 10 times.

3. A drunken friend claims he can predict the outcome of a fair coin flip. In 10 trials, he is correct all 10 times.

In all three cases, a classical test of $H_0 : \theta = 0.5$ rejects with significance level $2^{-10} \approx 0.001$.

Yet our conclusions differ sharply. We believe the music expert. We dismiss the drunk as a lucky streak. We are unsure about the tea lady. The data is the same; what differs is **prior information**. Any framework that ignores this gives the same answer to all three.

Inference is a special case of decision theory

Many statisticians treat "inference" as separate from decision making. Berger argues this is a mistake:

1. Statistical inferences should be constructed so they can be *used* in decision making. Many classical inferences fail at this.
2. The investigator often *does* have prior knowledge and information about consequences. Ignoring it is wasteful.
3. Inference itself *is* a decision problem: the action space is the set of possible inference statements, and the loss measures how well knowledge was conveyed.

Everything you learned in your first year – MLEs, Neyman–Pearson tests, confidence sets – can be reframed as special cases of decision theory. This course builds the framework underneath them.

The plan

This is a **decision theory** course, not a Bayesian analysis course. Bayesian tools enter when they serve the decision theory – to compute optimal rules, to prove admissibility, to find minimax procedures.

Now let us set up the formal language.

The ingredients

A **statistical decision problem** has five ingredients:

1. **State of nature** $\theta \in \Theta$: the unknown quantity. When experiments are performed to learn about θ, we call it the *parameter* and Θ the *parameter space*.
2. **Action space** $\mathcal{A}$: the set of possible decisions.
3. **Loss function** $L(\theta, a)$: the penalty for action a when the truth is θ. We require $L(\theta, a) \geq -K > -\infty$.
4. **Data** $X \in \mathcal{X}$, with distribution P_θ (density $f(x|\theta)$).
5. **Prior** $\pi(\theta)$: a probability distribution on Θ. (Not always available.)

Example 1: the drug company (Berger, Ex. 1)

Estimate θ_2 (market share) for an advertising campaign. Overestimating risks penalties for misleading advertising. Underestimating undersells the product.

- $\Theta = [0, 1], \quad \mathcal{A} = [0, 1] \quad$ (for estimation, typically $\mathcal{A} = \Theta$)
- Data: survey n people; $X \sim \mathrm{Bin}(n, \theta_2)$
- Loss: $L(\theta_2, a) = \begin{cases} \theta_2 - a & \text{if } \theta_2 \geq a \\ 2(a - \theta_2) & \text{if } \theta_2 < a \end{cases}$

  An overestimate is twice as costly as an underestimate.
- Prior: past drug launches suggest $\theta_2 \sim \mathrm{Uniform}(0.1, 0.2)$.

Berger notes: this development of L, f, and π is quite crude. Usually much more careful constructions are required. The techniques for doing this will be developed as we proceed.

Example 2: the transistor shipment (Berger, Ex. 2)

A radio company receives a shipment of transistors. Too expensive to check each one. A random sample of n is tested; $X \sim \mathrm{Bin}(n, \theta)$, where θ is the proportion defective.

- $\Theta = [0, 1], \quad \mathcal{A} = \{a_1, a_2\}$: accept or reject.
- $L(\theta, a_1) = 10\theta$: accepting a lot with defect rate θ produces defective radios.
- $L(\theta, a_2) = 1$: rejecting incurs a fixed cost (inconvenience, delay, replacement).
- The factor 10 encodes relative severity: a bad accept is much worse than a bad reject.
- Prior from past shipments: $\theta \sim \mathrm{Beta}(0.05, 1)$.

Example 3: the bond investment (Berger, Ex. 3)

An investor: buy risky ZZZ bonds (a_1) or a safe investment (a_2). **No data.**

$\Theta = \{\theta_1, \theta_2\}$ (no default vs. default), $\quad \mathcal{A} = \{a_1, a_2\}$.

When both Θ and $\mathcal{A}$ are finite, the loss is most easily represented as a **loss matrix**:

	a_1 (buy)	a_2 (safe)
θ_1 (no default)	-500	-300
θ_2 (default)	1000	-300

A gain is a negative loss. Prior: $\pi(\theta_1) = 0.9$, $\pi(\theta_2) = 0.1$.

This is a **no-data problem**: there is no statistical experiment. A decision rule is simply an action.

The loss function encodes consequences

The loss function $L(\theta, a)$ is the heart of the formulation. It forces you to think precisely about what matters.

We have already seen three different kinds:

- **Asymmetric linear loss** (drug company): overestimation and underestimation have different costs.
- **Action-dependent loss** (transistor shipment): one action has loss proportional to θ, the other has constant loss.
- **Loss matrix** (bond investment): when Θ and $\mathcal{A}$ are finite, L is a table.

The choice of loss is part of the modeling. Different losses lead to different optimal procedures. This is a feature, not a bug. We will study loss functions systematically in Lecture 2 (Berger Ch 2).

What should we do with the loss?

We have Θ, $\mathcal{A}$, $L(\theta, a)$, and data X.

Ideally, we would pick the action a that minimizes $L(\theta, a)$.

But we do not know θ.

For the bond problem: if $\theta = \theta_1$ (no default), we should buy ($L = -500 < -300$). If $\theta = \theta_2$ (default), we should not ($L = -300 < 1000$). The best action depends on the unknown state.

And when data X is available, the loss also depends on the random observation. We need a systematic way to handle both sources of uncertainty.

Decision rules

When data X is available, our action can depend on what we observe.

Definition

A (nonrandomized) **decision rule** is a function $\delta : \mathcal{X} \to \mathcal{A}$.

When $X = x$ is observed, the statistician takes action $\delta(x)$.

For a **no-data problem**, a decision rule is simply an action.

Examples:

- Drug company: $\delta(x) = x/n$ (estimate θ_2 by sample proportion).
- Transistor shipment: $\delta(x) = a_1$ if $x/n \leq 0.05$, else $\delta(x) = a_2$.

Given a rule δ, the loss incurred is $L(\theta, \delta(X))$. This depends on the unknown θ and the random X. We still cannot evaluate it directly. We need to average over something.

Averaging over X: the frequentist risk

For a fixed θ, we can average the loss over the randomness in X.

Definition

The **risk function** of a decision rule δ is

$$R(\theta, \delta) = E_{P_\theta}[L(\theta, \delta(X))] = \int_{\mathcal{X}} L(\theta, \delta(x))\, dP_\theta(x).$$

The risk answers: for each state of nature θ, how much loss would I expect on average if I used δ repeatedly with varying X?

$R(\theta, \delta)$ is a *function of* θ, not a number. Since θ is unknown, we have a problem in saying what "small" means. Different decision rules will have smaller risk at different values of θ.

Example 4: the normal mean (Berger, Ex. 4)

$X \sim \mathcal{N}(\theta, 1)$. Estimate θ under squared-error loss $L(\theta, a) = (\theta - a)^2$.

Consider the family of rules $\delta_c(x) = cx$. Then:

$$R(\theta, \delta_c) = E_{P_\theta}[(\theta - cX)^2] = c^2 + (1-c)^2\theta^2.$$

Some specific rules:

- $\delta_1(x) = x$: $R(\theta, \delta_1) = 1$ for all θ. Constant risk.
- $\delta_{1/2}(x) = x/2$: $R(\theta, \delta_{1/2}) = \frac{1}{4} + \frac{1}{4}\theta^2$. Smaller near $\theta = 0$, larger for $|\theta|$ big.
- $\delta_0(x) = 0$: $R(\theta, \delta_0) = \theta^2$. Great near zero, terrible far away.

The partial ordering of decision rules

Risk functions cross, so we cannot totally order decision rules. But we can define a **partial ordering**.

Definition

δ_1 is **R-better** than δ_2 if $R(\theta, \delta_1) \leq R(\theta, \delta_2)$ for all $\theta \in \Theta$, with strict inequality for some θ.

Definition

δ_1 is **R-equivalent** to δ_2 if $R(\theta, \delta_1) = R(\theta, \delta_2)$ for all θ.

Definition

δ is **admissible** if no R-better rule exists. It is **inadmissible** if an R-better rule exists.

It is clear that an inadmissible rule should not be used: there exists another rule that is at least as good everywhere and strictly better somewhere.

Example 4 continued: admissibility

For $\delta_c(x) = cx$ with $X \sim \mathcal{N}(\theta, 1)$ and squared-error loss:

$$R(\theta, \delta_c) = c^2 + (1-c)^2\theta^2.$$

- For $c > 1$: $R(\theta, \delta_1) = 1 \leq c^2 + (1-c)^2\theta^2 = R(\theta, \delta_c)$ for all θ, with strict inequality at $\theta = 0$. So δ_1 R-dominates δ_c, and δ_c is **inadmissible**.
- For $0 \leq c \leq 1$: risk functions cross. No rule dominates another. All are **admissible**.

But admissibility is a weak property. The rule $\delta_0(x) = 0$ – always estimate θ as zero, ignore the data – is admissible!

"This indicates that while admissibility may be a desirable property for a decision rule, it gives no assurance that the decision rule is reasonable." (Berger)

Example 5: a finite loss matrix (Berger, Ex. 5)

A no-data problem with three actions:

	a_1	a_2	a_3
θ_1	1	3	4
θ_2	-1	5	5
θ_3	0	-1	-1

- a_2 R-dominates a_3: $L(\theta_i, a_2) \leq L(\theta_i, a_3)$ for all i, strict at θ_1. So a_3 is inadmissible.
- a_1 and a_2: noncomparable. a_1 better at θ_1, θ_2; a_2 better at θ_3. Both admissible.

After eliminating inadmissible rules, we still have multiple admissible rules with crossing risk (or loss) functions. We need additional principles to choose among them.

A different way to average: over θ

The frequentist averaged the loss over X (for each θ), producing the risk function $R(\theta, \delta)$. But since θ is unknown, the risk is a function we cannot evaluate.

The Bayesian takes a different route: if we have a prior π^* for θ, we can average the loss over θ directly.

Definition

The **Bayesian expected loss** of an action a is

$$\rho(\pi^*, a) = E_{\pi^*}[L(\theta, a)] = \int_\Theta L(\theta, a)\, d\pi^*(\theta).$$

This produces a single number for each action. Comparison is straightforward: pick the a with smallest $\rho(\pi^*, a)$. Such an action is called a **Bayes action**, denoted a^{π^*}.

Example 3 revisited: the bond investment

$$\rho(\pi, a_1) = L(\theta_1, a_1)\pi(\theta_1) + L(\theta_2, a_1)\pi(\theta_2) = (-500)(0.9) + (1000)(0.1) = -350.$$

$$\rho(\pi, a_2) = (-300)(0.9) + (-300)(0.1) = -300.$$

Since $-350 < -300$, the Bayes action is a_1: buy the bonds.

For Example 1 (drug company), one can similarly compute $\rho(\pi, a)$ as a function of a for $\pi = \mathrm{Uniform}(0.1, 0.2)$ and find the Bayes action a^π.

Bayes risk of a decision rule (Berger §1.3.2)

We can also average the *risk function* over θ using a prior. This evaluates entire decision rules, not just single actions.

Definition

The **Bayes risk** of a decision rule δ with respect to a prior π is

$$r(\pi, \delta) = E_\pi[R(\theta, \delta)] = \int_\Theta R(\theta, \delta)\, d\pi(\theta).$$

A **Bayes rule** δ^π minimizes $r(\pi, \delta)$ over all rules δ. The quantity $r(\pi) = r(\pi, \delta^\pi)$ is the **Bayes risk** for π.

The Bayes risk collapses the entire risk function – a function of θ – into a single number. This solves the comparison problem, at the cost of requiring a prior.

Example 4 continued: the Bayes rule and shrinkage

$X \sim \mathcal{N}(\theta, 1)$, $L = (\theta - a)^2$, $\delta_c(x) = cx$. Suppose $\pi(\theta) = \mathcal{N}(0, \tau^2)$.

$$r(\pi, \delta_c) = E_\pi[R(\theta, \delta_c)] = E_\pi[c^2 + (1-c)^2\theta^2] = c^2 + (1-c)^2\tau^2.$$

Minimizing over c: $\quad c_0 = \dfrac{\tau^2}{1+\tau^2}$.

The Bayes rule is $\delta^\pi(x) = \dfrac{\tau^2}{1+\tau^2}\,x$. It **shrinks** x toward zero.

Large τ^2 (diffuse prior, little information about θ): $c_0 \approx 1$, almost no shrinkage. Small τ^2 (concentrated prior, strong belief $\theta \approx 0$): $c_0 \approx 0$, heavy shrinkage.

This is our first encounter with shrinkage. It will return in Week 6 when we study the Stein phenomenon.

Where we stand

We have set up the decision problem $(\Theta, \mathcal{A}, L, X, \pi)$, defined decision rules $\delta : \mathcal{X} \to \mathcal{A}$, and introduced two ways to evaluate them: the frequentist risk $R(\theta, \delta)$ and the Bayesian expected loss $\rho(\pi, a)$ / Bayes risk $r(\pi, \delta)$.

So far, every decision rule deterministically maps an observation to an action. There is one more generalization we need before the framework is complete: what happens if we allow the statistician to **randomize**?

Motivation: matching pennies (Berger, Ex. 6)

You and an opponent simultaneously uncover a penny. If the coins match, you win \$1. If they do not match, you lose \$1.

$\Theta = \{\theta_1, \theta_2\}$ (opponent's coin), $\mathcal{A} = \{a_1, a_2\}$ (your coin). Loss matrix:

	a_1 (heads)	a_2 (tails)
θ_1 (opp. heads)	-1	1
θ_2 (opp. tails)	1	-1

Both a_1 and a_2 are admissible. But if the game is played repeatedly, committing to either action exclusively is disastrous: the opponent will notice and exploit it. Any patterned choice can be detected. The only way to prevent this is to choose a_1 and a_2 **randomly**.

Definition of a randomized decision rule

Definition

A **randomized decision rule** $\delta^*(x, \cdot)$ is, for each x, a probability distribution on $\mathcal{A}$. If x is observed, $\delta^*(x, A)$ is the probability that an action in $A \subseteq \mathcal{A}$ will be chosen.

In no-data problems, a randomized rule is simply a probability distribution on $\mathcal{A}$, called a **randomized action**.

Nonrandomized rules are a special case: $\delta(x)$ corresponds to the randomized rule that puts probability one on the action $\delta(x)$.

For matching pennies: $\delta^*(a_1) = p$, $\delta^*(a_2) = 1 - p$.

Loss and risk of randomized rules

The loss of a randomized rule is defined by averaging over the randomization:

Definition

$$L(\theta, \delta^*(x, \cdot)) = E_{\delta^*(x,\cdot)}[L(\theta, a)],$$

where the expectation is over a drawn from $\delta^*(x, \cdot)$. The risk is then

$$R(\theta, \delta^*) = E_{P_\theta}[L(\theta, \delta^*(X, \cdot))].$$

For matching pennies:

$$L(\theta_1, \delta^*) = p(-1) + (1-p)(1) = 1 - 2p, \quad L(\theta_2, \delta^*) = p(1) + (1-p)(-1) = 2p - 1.$$

If $p = \frac{1}{2}$, the loss is zero no matter what the opponent does. The randomized rule with $p = \frac{1}{2}$ guarantees an expected loss of zero.

Example 7: hypothesis testing with discrete data (Berger, Ex. 7)

Observe $X \sim \mathrm{Bin}(n, \theta)$. Test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ (with $\theta_0 > \theta_1$).

Actions: a_0 = accept H_0, a_1 = reject H_0. Loss: 0 for a correct decision, 1 for incorrect.

The most powerful nonrandomized tests have rejection regions $C = \{x : x \leq j\}$ for $j = 0, 1, \ldots, n$. The size of such a test is $\alpha = P_{\theta_0}(X \leq j)$.

The problem: since X is discrete, these sizes can only take finitely many values. For example, with $n = 10$ and $\theta_0 = 0.5$:

j	0	1	2	3	4
$\alpha = P_{\theta_0}(X \leq j)$	0.001	0.011	0.055	0.172	0.377

If you want a test of size exactly $\alpha = 0.05$, no nonrandomized test achieves it.

Example 7 continued: randomization solves the problem

Consider the randomized rule: reject H_0 with probability 1 if $x < j$, with probability p if $x = j$, and with probability 0 if $x > j$:

$$\delta_j^*(x, a_1) = \begin{cases} 1 & \text{if } x < j \\ p & \text{if } x = j \\ 0 & \text{if } x > j \end{cases}$$

The risk under θ_0 (the size) is:

$$R(\theta_0, \delta_j^*) = P_{\theta_0}(X < j) + p\, P_{\theta_0}(X = j).$$

By choosing j and p appropriately, we can achieve **any** desired size $\alpha \in [0, 1]$.

This is exactly what happens in the Neyman–Pearson lemma for discrete distributions: the most powerful test at a given size may require randomization at the boundary.

This is not a game against an adversary. It is a genuine statistical reason to allow randomized rules.

Should we actually randomize?

Matching pennies involves an intelligent adversary. Most statistical problems do not. In the absence of an adversary, leaving the final choice of action to a coin flip seems hard to justify.

Berger is direct: "We will rarely recommend the actual use of a randomized rule."

Then why study them? We have already seen two reasons:

1. **Exact size in discrete testing** (Example 7): without randomization, certain size levels are simply unattainable.
2. **Intelligent adversaries** (Example 6): the minimax theorem in Chapter 5 requires mixed strategies, i.e., randomized rules.

A third, more structural reason: the class of nonrandomized rules may not be convex. Randomization convexifies it. Many key theorems in decision theory (minimax, complete class) require this convexity.

Three principles: a preview (Berger §1.5)

We now have two ways to evaluate decision rules: $R(\theta, \delta)$ and $r(\pi, \delta)$. These lead to three selection principles:

I. Bayes risk principle: Choose δ to minimize $r(\pi, \delta)$.

Requires a prior. Gives a unique answer (usually).

II. Minimax principle: Choose δ to minimize $\sup_\theta R(\theta, \delta)$.

No prior needed. Protects against the worst case.

III. Invariance principle: If the problem has a symmetry, restrict to rules that respect it.

A central theme of this course: these three are deeply connected. We begin to explore these connections in Lecture 2.

The questions that drive this course

1. When is a Bayes rule also minimax? (Weeks 3–5)
2. When is the "standard" estimator inadmissible? (Week 6)
3. When does symmetry pin down the optimal rule? (Weeks 7–8)
4. When should you stop collecting data? (Weeks 8–9)
5. Can you characterize *all* admissible rules? (Week 9)
6. How do these ideas reappear in modern machine learning? (Week 10)